SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Data Warehousing and Data Mining

COURSE CODE: CSA1609

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding ***(2)***	Method Selection ***(2)***	Solution Process ***(2.5)***	Accuracy of Calculation ***(2)***	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I, G Madhulatha(192311295) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G Madhulatha

Annexure B
Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the **annual premium increase rate (xi)** and the **number of claims per 100 policyholders (yi)** in a health insurance company.

Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.
Before computing covariance, calculate the **mean of x and y**.

Premium Increase % (xi)	Claims per 100 Policyholders (yi)
3.2	15
4.0	18
5.5	25
4.8	20

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2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analyzing **patient blood pressure readings (systolic)** to identify trends. Due to minor fluctuations, the analyst performs **local smoothing using equal-frequency (equi-depth) binning**.

Sample Data Set (Systolic BP readings):
110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

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3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification.
To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**.

Sample Data Set (Sorted BMI):
18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

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4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**.
To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.

Sample Data Set (Costs in ₹):
120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

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5. Standardization of Patient Age Data (5 Marks)

Given the following **ages of insured individuals**:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:

(a) Compute the **mean absolute deviation (MAD)** of age.
(b) Compute the **z-score for the first four observations**.

1. COVARIANCE ANALYSIS IN HEALTH INSURANCE

The table below shows the annual premium increase rate (xi) and the number of claims per 100 policyholders (yi) in a health insurance company. Using the covariance formula, determine whether premium increase and claim frequency have a positive or inverse relationship. Before computing covariance, calculate the mean of x and y.

Data:

Premium Increase % (xi)	Claims per 100 Policyholders (yi)
3.2	15
4.0	18
5.5	25
4.8	20

Step 1: Calculate Mean of X

$\bar{x} = (3.2 + 4.0 + 5.5 + 4.8) / 4$
$\bar{x} = 17.5 / 4$
$\bar{x} = 4.375$

Therefore, mean of X = 4.375.

Step 2: Calculate Mean of Y

$\bar{y} = (15 + 18 + 25 + 20) / 4$
$\bar{y} = 78 / 4$
$\bar{y} = 19.5$

Therefore, mean of Y = 19.5.

Step 3: Calculate Covariance

Formula:

$\mathrm{Cov}(X,Y) = \Sigma[(xi - \bar{x})(yi - \bar{y})] / n$

Table:

xi	yi	$xi - \bar{x}$	$yi - \bar{y}$	$(xi - \bar{x})(yi - \bar{y})$
3.2	15	-1.175	-4.5	5.2875
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125

Total product = 12.25

Cov(X,Y) = 12.25 / 4
Cov(X,Y) = 3.0625

FINAL ANSWER:
Covariance = 3.0625.

Since the covariance is positive, premium increase and claim frequency have a positive/direct relationship.

CONCLUSION:
When the premium increase rate increases, the number of claims tends to increase as well.

2. DATA SMOOTHING USING EQUAL-FREQUENCY BINNING

A hospital is analyzing patient blood pressure readings (systolic) to identify trends. Due to minor fluctuations, the analyst performs local smoothing using equal-frequency (equi-depth) binning.

Sample Data Set:

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

There are 27 observations.

For 3 equal-frequency bins:

27 / 3 = 9 observations per bin.

BIN 1:
110, 112, 115, 115, 118, 120, 120, 122, 124

Mean = 1056 / 9 = 117.33

Smoothed Bin 1:
117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33

BIN 2:
124, 126, 126, 126, 126, 130, 135, 135, 138

Mean = 1166 / 9 = 129.56

Smoothed Bin 2:
129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56

BIN 3:
138, 138, 138, 140, 145, 150, 152, 160, 180

Mean = 1341 / 9 = 149

Smoothed Bin 3:
149, 149, 149, 149, 149, 149, 149, 149, 149

FINAL ANSWER:
The smoothed data consists of:
117.33 repeated 9 times,
129.56 repeated 9 times,
149 repeated 9 times.

3. EQUAL-WIDTH BINNING FOR RISK CATEGORIZATION

An insurance company evaluates Body Mass Index (BMI) of 18 clients for risk classification. To simplify analysis, they smooth the data by replacing values with closest bin boundaries using equal-width partitioning.

Sample Data Set:

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Minimum BMI = 18.5
Maximum BMI = 39.5

Assuming 3 equal-width bins:

Bin width = (39.5 − 18.5) / 3
Bin width = 21 / 3
Bin width = 7

Therefore:

Bin 1: 18.5 – 25.5
Bin 2: 25.5 – 32.5
Bin 3: 32.5 – 39.5

Replace each value with the closest bin boundary.

BIN 1:
18.5 → 18.5
19.2 → 18.5
21.8 → 18.5
24.9 → 25.5

BIN 2:
25.5 → 25.5
26.2 → 25.5

26.4 → 25.5
27.8 → 25.5
29.2 → 32.5
30.2 → 32.5
30.4 → 32.5
31.9 → 32.5
32.4 → 32.5

BIN 3:
33.1 → 32.5
33.6 → 32.5
34.7 → 32.5
38.2 → 39.5
39.5 → 39.5

FINAL SMOOTHED DATA:

18.5, 18.5, 18.5, 25.5,
25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5,
32.5, 32.5, 32.5, 39.5, 39.5

4. DATA REDUCTION USING BIN MEDIANS

A hospital administration is reviewing the cost of medical consumables (in ₹). To reduce data for mining purposes, they discretize the sorted price records using bin medians.

Original Data:

120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

First sort the data:

120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

There are 27 records.

For 3 equal-frequency bins:

27 / 3 = 9 observations per bin.

BIN 1:
120, 140, 145, 145, 148, 150, 150, 150, 155

Median = 148

BIN 2:
155, 155, 155, 160, 160, 165, 170, 170, 175

Median = 160

BIN 3:
175, 175, 175, 180, 190, 200, 210, 230, 300

Median = 190

FINAL REDUCED DATA:

148, 148, 148, 148, 148, 148, 148, 148, 148,
160, 160, 160, 160, 160, 160, 160, 160, 160,
190, 190, 190, 190, 190, 190, 190, 190, 190

CONCLUSION:
The 27 original values are represented using only 3 median values: 148, 160, and 190.

5. STANDARDIZATION OF PATIENT AGE DATA

Given ages:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

(a) COMPUTE MEAN ABSOLUTE DEVIATION (MAD)

Step 1: Calculate Mean

$\bar{x} = (18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28) / 10$

$\bar{x} = 330 / 10$
$\bar{x} = 33$

Therefore, Mean = 33.

Step 2: Calculate Absolute Deviations

Age | |Age − 33|
18 | 15
22 | 11
25 | 8
42 | 9
28 | 5
43 | 10
33 | 0
35 | 2
56 | 23
28 | 5

Sum of absolute deviations = 88

MAD = 88 / 10
MAD = 8.8

FINAL ANSWER:
MAD = 8.8

(b) COMPUTE Z-SCORE FOR FIRST FOUR OBSERVATIONS

Formula:

$z = (x - \bar{x}) / \sigma$

Mean = 33

Population standard deviation:

$\sigma = \sqrt{[\Sigma(x - 33)^2 / 10]}$

$\sigma \approx 10.835$

For age 18:

$z = (18 - 33) / 10.835$
$z = -15 / 10.835$
$z = -1.384$

For age 22:

$z = (22 - 33) / 10.835$
$z = -11 / 10.835$
$z = -1.015$

For age 25:

$z = (25 - 33) / 10.835$
$z = -8 / 10.835$
$z = -0.738$

For age 42:

$z = (42 - 33) / 10.835$
$z = 9 / 10.835$
$z = 0.831$

FINAL Z-SCORE TABLE:

Age | Z-score
18 | -1.384

22 | -1.015
25 | -0.738
42 | 0.831

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation